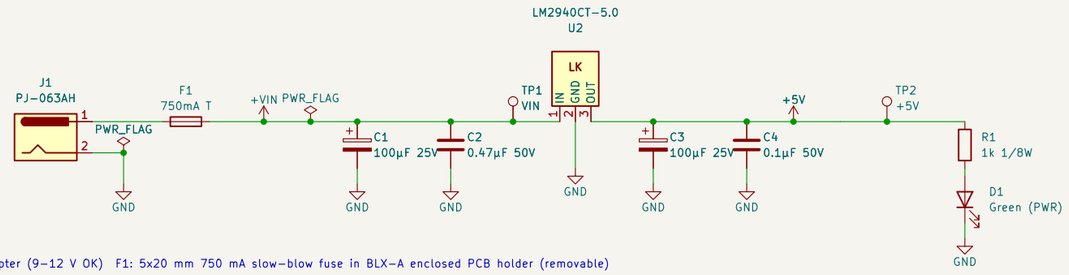
## POWER INPUT, PROTECTION & 5 V REGULATION



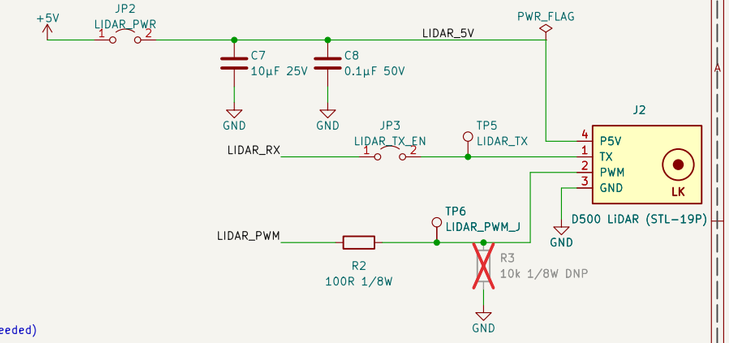

J1: 9 V DC adapter (9–12 V OK)  F1: 5x20 mm 750 mA slow-blow fuse in BLX-A enclosed PCB holder (removable)

Each board makes its own 5 V from its own 9 V supply; only GND is shared on the ribbon (pin 8).

U2: 1 A LDO, 1 V max dropout, reverse-battery protected; C2 / C3 per datasheet (0.47 µF in, 100 µF low-ESR out)

U2 dissipates ~2.0 W at 9 V in (~3.9 W at 12.6 V): heatsink fitted. D1 = power-on indicator.

## LIDAR INTERFACE – WAVESHARE D500 (JST PH 2.0 mm, 4-pin)



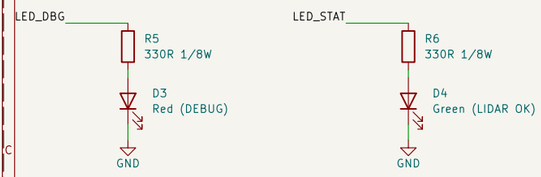

D500: 5 V / 290 mA, UART 230400 8N1, 3.3 V logic (no level shifter needed)

R3 NOT FITTED: D500 needs PWM floating for its internal 10 Hz speed control (fit R3 only for an LD19)

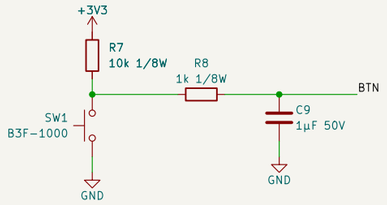
JP2 isolates LiDAR power, JP3 isolates LiDAR TX (sniff / inject data)

RC6 stays a high-Z input unless firmware drives PWM; R2 limits current into PWM

## DEBUG LEDs & PUSHBUTTON




LED current: (3.3 – 2.0) / 330 = ~4 mA

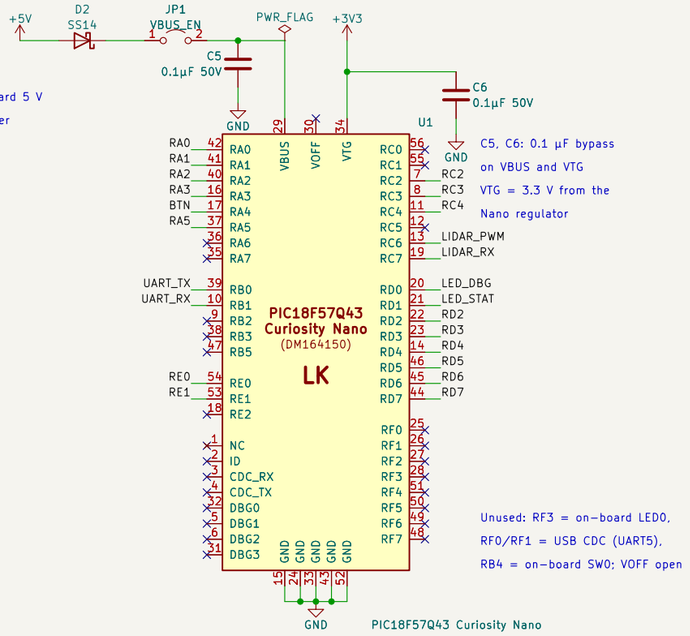



SW1 active-low. Hardware debounce:

(R7 + R8) x C9 = 11 ms release, R8 x C9 = 1 ms press

BTN pin uses Schmitt-trigger input (INLVL)

## MICROCONTROLLER – PIC18F57Q43 CURIOSITY NANO




D2 + JP1: USB 5 V and board 5 V cannot back-feed each other

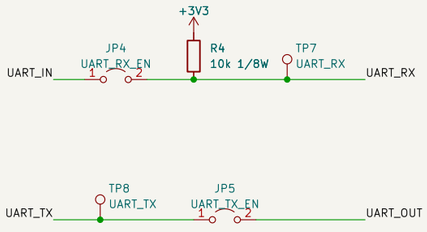
C5, C6: 0.1 µF bypass on VBUS and VTG

VTG = 3.3 V from the Nano regulator

Unused: RF3 = on-board LED0, RF0/RF1 = USB CDC (UART5), RB4 = on-board SW0; VOFF open

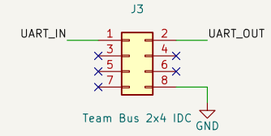
PIC18F57Q43 Curiosity Nano

## TEAM BUS – RIBBON CABLE TO MASTER




R4: pull-up keeps RX idle-high if the ribbon is unplugged

JP4 / JP5 isolate the team bus for debugging

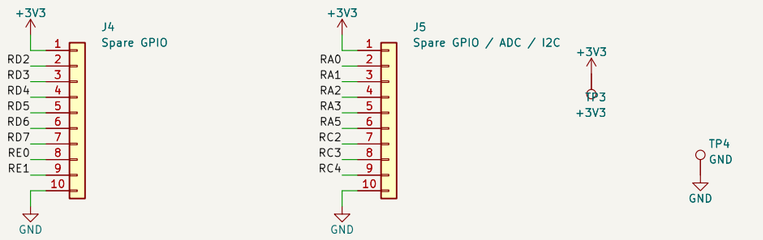



J3 pin 1 = UART in (master TX -> RB1 / U2RX)

J3 pin 2 = UART out (RB0 / U2TX -> master RX)

J3 pins 3–5 digital, 6–7 analog (unused here), pin 8 = GND

## EXPANSION HEADERS & TEST POINTS




Other test points: TP1 VIN, TP2 +5V, TP5–TP8 on LiDAR / UART signals

## DESIGN NOTES

LiDAR subsystem of OpenField (Team 1): reads the D500 360° DTOF LiDAR on UART1 (230400 baud) and reports to the MASTER board on UART2 via the ribbon.

Pin map: RC7 LIDAR_RX (U1RX), RC6 LIDAR_PWM, RB1 UART_RX (U2RX), RB0 UART_TX (U2TX), RD0 LED_DBG, RD1 LED_STAT, RA4 BTN

Power: +5V load 402 mA (LiDAR 290 + Nano 100 + LEDs 12) -> 502.5 mA with 25% margin; LM2940 rated 1 A -> 497.5 mA headroom

Ratings: all resistors 0805 1/8 W (>= 50% margin); capacitor voltages >= 25% above the rail they sit on

Custom symbols with initials (LK): U1 Curiosity Nano, J2 D500 LiDAR, U2 VReg

Analog sensor / actuator requirement is met at the system level (wheel encoders, cliff sensors), per the instructor.

Waveshare D500 LiDAR + PIC18F57Q43 Curiosity Nano

Landon Kuhman (LK)

**RAS 304 Embedded Systems – Team 1 (OpenField)**

Sheet: /

File: LiDAR_Subsystem_LK.kicad_sch

***Title: LiDAR Subsystem Schematic***

| Size: A3 | Date: 2026-11-02 | **Rev: 1.1** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |